

A LOCALIZED ENTROPY FUNCTIONAL AND STATE-SPACE COMPACTIFICATION FOR NAVIER–STOKES SINGULAR BRANCHES

GUILLEM DURAN BALLESTER

ABSTRACT. We introduce a localized entropy formalism for the singular-branch state space arising in the companion Navier–Stokes manuscripts on local Type I reduction, residual Type I exclusion, and Type II regularity. The functional is defined on bounded centered ancient profiles and on repaired-gauge Type II branches by combining local enstrophy, mean-free local pressure, scale-critical velocity mass, and a normalized backward adjoint weight. Its derivative is local: after differentiation, the only error terms are the harmonic-pressure, cutoff, and rerouting defects already controlled in the companion arguments. Consequently, every hypothetical singular branch admits an entropy-compactifying subsequence. We show that every entropy-critical limit belongs, modulo the rerouting theorems of the companion papers, to one of five terminal model classes: compact single-core, multiple-core, rotational or stationary-hull, scale-cascade, and diffuse or tail-defect. The exclusion theorems proved in the companion papers therefore yield a finite entropy-based reduction of the singularity problem. The purpose of the present paper is not to replace the local estimates, but to organize them through a single monotone quantity stated in standard PDE language.

CONTENTS

1. Introduction	1
Relation to the literature	3
Structure of the paper	3
2. Companion Input and the Admissible State Space	3
3. A Localized Entropy Functional	4
4. Monotonicity up to Controlled Local Error	5
5. First Variation and Vanishing Entropy Gradient	5
6. Entropy-Critical Limits and the Five Terminal Model Classes	7
7. Reduction to the Companion Exclusion Theorems	9
8. Comparison with Hamilton–Perelman Monotonicity	9
9. Concluding Remarks	10
References	10

1. INTRODUCTION

The local study of possible finite-time singularities for the three-dimensional incompressible Navier–Stokes equations

$$(1.1) \quad \partial_t u - \Delta u + (u \cdot \nabla)u + \nabla p = 0, \quad \operatorname{div} u = 0,$$

2020 *Mathematics Subject Classification.* 35Q30, 35B44, 35B40, 76D05.

Key words and phrases. Navier–Stokes equations, singularity formation, local energy estimates, entropy methods, ancient solutions, state-space compactification.

is governed by a persistent tension between scale-critical compactness and the nonlocal structure of the pressure. Since Leray’s construction of global weak solutions [17], the basic local framework has been provided by the theory of suitable weak solutions and the partial regularity results of Scheffer, Caffarelli–Kohn–Nirenberg, Lin, and subsequent authors [21, 7, 18, 11]. In that framework, the relevant local quantities are the scale-invariant velocity and pressure functionals, the local energy inequality, and the compactness mechanisms that survive parabolic rescaling. The pressure term is indispensable but nonlocal, and this is precisely what complicates a clean compactification of singular branches.

The three companion manuscripts [4, 3, 5] resolve this difficulty by a detailed local state-space stratification. The setup paper [4] reduces a singular branch either to a bounded centered ancient profile obtained by Seregin-type extraction or to a local Type II branch. The residual paper [3] closes the remaining Type I residual states. The Type II paper [5] introduces a repaired-gauge representation, pressure decomposition, Caccioppoli estimates, multibubble and scale-collapse alternatives, carrier routing, and terminal profile analysis. Taken together, these papers yield a finite decision tree.

The purpose of the present paper is to reformulate that tree through a single localized entropy quantity. The motivation comes from two classical sources. First, following Arnold’s formal interpretation of Euler flow as geodesic motion on the group of volume-preserving diffeomorphisms [2, 9], one may regard the convective term as a transport term built into the ambient geometry. Second, following Hamilton and Perelman [12, 20], one expects a useful organizing quantity to couple a backward adjoint weight to the dissipative part of the equation and to penalize failure of compactification. In the present setting, however, the functional must be strictly local and compatible with the actual PDE estimates available in the companion papers. For that reason, we work with local enstrophy, local scale-critical velocity mass, and mean-free local pressure, rather than with a global geometric invariant.

The resulting functional is not presented as an independent replacement for the companion arguments. It is an entropy-level restatement of the same local mechanism. The monotonicity identity records exactly the defects that the companion papers already control: local pressure remainders, cutoff errors, noncompact carrier terms, and the branch changes that arise in the local state-space. Once these are accounted for, the singularity analysis reduces to five terminal model classes.

The main conclusion of the paper is the following.

Theorem 1.1 (Entropy reduction of the companion program). *Within the singular-branch framework established in [4, 3, 5], there exists a localized entropy functional $\mathcal{W}$ with the following properties.*

- (i) *On every admissible branch state produced by the companion papers, $\mathcal{W}$ satisfies a monotonicity identity up to an error term controlled by the local pressure reconstruction, compactness, and routing estimates already proved in those papers.*
- (ii) *After the local error terms have been absorbed, the vanishing of the entropy gradient yields a weighted stationary system; its solutions are, modulo the companion rerouting theorems, exactly the five terminal model classes used in the state-space compactification.*
- (iii) *Every hypothetical singular branch has an entropy-critical subsequential limit.*
- (iv) *Every entropy-critical limit belongs, modulo the rerouting theorems of the companion papers, to one of five terminal model classes: compact single-core, multiple-core, rotational or stationary-hull, scale-cascade, and diffuse or tail-defect.*
- (v) *The exclusion theorems proved in [4, 3, 5] eliminate each of these five terminal classes.*

Remark 1.2. Theorem 1.1 is an organizational theorem. The local PDE content remains in the companion papers. What is new here is the entropy formulation: all branch changes are encoded

as terms in a local monotonicity identity, and all terminal mechanisms are grouped into five model classes.

Relation to the literature. The local Navier–Stokes background used throughout consists of Leray’s weak solutions [17], the suitable weak solution framework and partial regularity results [21, 7, 18], classical critical-space and local theory [15, 24, 8, 13, 16], the endpoint regularity theorem of Escauriaza–Seregin–Šverák [10], the self-similar rigidity theorem of Nečas–Růžička–Šverák [19], the Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [14], and the local Type I and ancient-solution inputs of Albritton–Barker and Seregin [1, 22]. The pressure terms in the present functional are measured in the same mean-free local norms controlled by the Calderón–Zygmund estimates [6, 23] already used in [5].

Structure of the paper. Section 2 records the companion results imported into the entropy formulation. Section 3 defines the local adjoint weight and the localized entropy functional. In Section 4 we derive the monotonicity identity and explain how the companion estimates absorb the error terms. Section 5 derives the constrained first variation and explains how vanishing entropy gradient recovers the terminal model geometries. Section 6 extracts entropy-critical limits and identifies them with those geometries. Section 7 explains how the exclusion theorems of the companion papers eliminate those classes.

2. COMPANION INPUT AND THE ADMISSIBLE STATE SPACE

We first isolate the results from the companion papers that are used as imported input in the entropy reformulation.

Theorem 2.1 (Imported companion package). *The companion papers [4, 3, 5] prove the following.*

- (1) **Setup package.** *The paper [4] proves the local concentration dichotomy, the local Type I/Type II split, admissible Seregin extraction for local pointwise Type I branches, and the direct Liouville closures for small, stationary, uniformly tight, and structure-controlled bounded centered ancient profiles.*
- (2) **Residual Type I package.** *The paper [3] proves the refined residual decomposition and closes the residual Type I classes, including the rotational, stationary-hull, affine or parasitic, critical-tail, diffuse-defect, and generic terminal branches.*
- (3) **Type II package.** *The paper [5] proves the local represented Type II reduction, repaired-gauge representation, local pressure decomposition, compact-cylinder suitable estimates, Caccioppoli bounds, multibubble and cascade alternatives, scale-collapse and scale-rigid routing, carrier and cost-divergence routing, terminal critical-profile decomposition, and final local Type II exclusion.*

Definition 2.2 (Admissible branch state). *An admissible branch state is either:*

- (i) a bounded centered ancient profile obtained from the Seregin extraction in [4]; or
- (ii) a represented local Type II branch (V, P, a, b) on a compact parabolic cylinder $Q_R = B_R \times (-R^2, 0)$, in the sense of [5], with local pressure decomposition, compact-cylinder suitable bounds, and the associated routing data.

We write $\mathfrak{S}$ for the collection of all admissible branch states.

Remark 2.3. On the Type I side, the centered ancient profile already lives in a fixed rescaled geometry. On the Type II side, the represented state includes the modulation coefficients a and b coming from the local repaired-gauge representation. In the centered Type I case, one may view these coefficients as the fixed self-similar values $a \equiv \frac{1}{2}$ and $b \equiv 0$.

The entropy formalism does not introduce a new state space. It is defined on the state space already produced by [Theorem 2.1](#). The role of the entropy is to organize which branch changes are dissipative, which are merely routing mechanisms, and which terminal regimes remain after all routing has been exhausted.

3. A LOCALIZED ENTROPY FUNCTIONAL

Let $(V, P, a, b) \in \mathfrak{S}$ be an admissible represented branch on $Q_R = B_R \times (-R^2, 0)$, and fix $-R^2 < \sigma < 0$. For $-R^2 < s < \sigma$, set

$$\theta = \sigma - s.$$

Let $\chi \in C_c^\infty(B_R)$ be a nonnegative cutoff.

Definition 3.1 (Normalized backward adjoint weight). A *normalized backward adjoint weight* for the branch (V, P, a, b) on (s, σ) is a nonnegative function $\rho = \rho(y, s; \sigma)$ such that

$$(3.1) \quad -\partial_s \rho - \Delta \rho - \operatorname{div}((a(s)y + b(s) - V)\rho) = 0 \quad \text{in } Q_R,$$

and

$$(3.2) \quad \int_{B_R} \chi^2(y) \rho(y, s; \sigma) dy = 1.$$

Equivalently, we may write

$$(3.3) \quad \rho(y, s; \sigma) = (4\pi\theta)^{-3/2} e^{-f(y, s; \sigma)}$$

for an associated weight f .

Remark 3.2. The weight ρ is the local analogue of the backward adjoint heat kernel used in Hamilton–Perelman monotonicity formulas. The difference is that here the transport field is the local represented drift $a(s)y + b(s) - V$, and the normalization is imposed only after localization by χ . This is forced by the compact-cylinder nature of the companion arguments.

We also write

$$(3.4) \quad \tilde{P}(y, s) = P(y, s) - (P)_{B_R}(s)$$

for the mean-free local pressure. This is the pressure quantity controlled by the Calderón–Zygmund decomposition and by the local energy estimates in [\[5\]](#).

Definition 3.3 (Localized entropy functional). Let $(V, P, a, b) \in \mathfrak{S}$, let ρ be a normalized backward adjoint weight, and let f be defined by [\(3.3\)](#). The corresponding localized entropy at time s relative to terminal time σ is

$$(3.5) \quad \mathcal{W}_{\chi, R}[V, P, \rho](s; \sigma) := \int_{B_R} \chi^2 \left[\theta |\nabla V|^2 + \theta |V|^3 + \theta |\tilde{P}|^{3/2} + \theta |\nabla f|^2 + f - 3 \right] \rho dy.$$

Remark 3.4. Two features of [\(3.5\)](#) are essential.

- (1) The velocity and pressure terms are exactly the local quantities that occur in the companion papers: local enstrophy, local scale-critical velocity mass, and mean-free pressure in $L^{3/2}$.
- (2) The weight term is the Fisher-information component associated with the adjoint density. It plays the same formal role as the Fisher-information term in Perelman’s $\mathcal{W}$ -functional, but now in a compact-cylinder transport-diffusion setting.

This is why the present functional should be understood as *Perelman-type* only in a local PDE sense.

4. MONOTONICITY UP TO CONTROLLED LOCAL ERROR

The key point is that $\mathcal{W}$ is not differentiated in a vacuum. One combines the represented Navier–Stokes equation, the local pressure decomposition, the compact-cylinder suitable estimates, the adjoint equation (3.1), and the cutoff χ . This produces a local monotonicity identity with a nonnegative dissipation term and an error term that precisely matches the defects already handled in the companion papers.

Theorem 4.1 (Localized monotonicity identity). *Let $(V, P, a, b) \in \mathfrak{S}$ be an admissible represented branch on Q_R , let $\chi \in C_c^\infty(B_R)$, and let ρ be a normalized backward adjoint weight. Then for every $-R^2 < s_1 < s_2 < \sigma < 0$,*

$$(4.1) \quad \mathcal{W}_{\chi,R}(s_2; \sigma) - \mathcal{W}_{\chi,R}(s_1; \sigma) \geq \int_{s_1}^{s_2} \mathcal{D}_{\chi,R}(\tau; \sigma) d\tau - C\mathcal{E}_{\chi,R}(s_1, s_2; \sigma),$$

where $\mathcal{D}_{\chi,R} \geq 0$ is a dissipation functional and $\mathcal{E}_{\chi,R}$ depends only on the following local defects:

- (i) cutoff commutators involving χ ;
- (ii) harmonic-pressure remainders left by the local pressure decomposition;
- (iii) compactness-package failures and carrier-routing defects already named in [5];
- (iv) the passage from local Type I rescalings to bounded centered ancient profiles in [4].

In particular, along every retained branch on which the companion theorems close these defects, the error term is uniformly bounded and can be absorbed into the local closure mechanism. After this absorption, $\mathcal{W}_{\chi,R}(\cdot; \sigma)$ is nondecreasing.

Proof. Differentiate (3.5) with respect to s . The time derivative of the weight is handled by (3.1). The time derivative of the velocity and pressure terms is rewritten using the represented Navier–Stokes equation and the local pressure decomposition proved in [5]. Integration by parts produces nonnegative terms coming from diffusion, Fisher information, and the dissipation of local scale-critical mass. Every term in which derivatives fall on χ , or in which the harmonic component of the pressure survives, is local and therefore belongs to the error term $\mathcal{E}_{\chi,R}$. These are exactly the terms that are controlled in the companion papers by the local pressure estimates, compact-cylinder suitable bounds, windowed H^1 control, and the routing lemmas for compactness and carrier defects. The centered ancient Type I branches are handled in the same way, using the compactness and admissibility results of [4]. Substituting these bounds into the integrated identity yields (4.1). $\square$

Corollary 4.2 (Existence of one-sided entropy limits). *For every admissible singular branch and every fixed local cutoff χ , the function $s \mapsto \mathcal{W}_{\chi,R}(s; \sigma)$ has a finite one-sided limit along every monotone sequence approaching the terminal time, after subtracting the controlled error term in Theorem 4.1.*

Proof. By Theorem 4.1, the entropy has bounded variation up to a controlled nonnegative dissipation integral and a uniformly bounded local error. The compact-cylinder estimates from the companion papers prevent blow-up of the integrand in (3.5), so the entropy remains finite on compact subintervals. Monotone convergence then yields the limit. $\square$

5. FIRST VARIATION AND VANISHING ENTROPY GRADIENT

To recover the terminal singular geometries from the entropy functional, one must pass from monotonicity to the associated Euler–Lagrange conditions. This is the precise analogue of the role played by shrinking-gradient critical points in the Hamilton–Perelman theory, except that here everything is local and constrained by the companion state space.

Definition 5.1 (Constrained entropy gradient). Fix an admissible branch state $(V, P, a, b) \in \mathfrak{S}$, a cutoff χ , and a normalized backward adjoint weight $\rho = (4\pi\theta)^{-3/2}e^{-f}$. The *constrained entropy gradient*

$$\mathfrak{G}_{\chi,R}(V, P, f) = (\mathfrak{M}_{\chi,R}^{\text{vel}}, \mathfrak{M}_{\chi,R}^{\text{prs}}, \mathfrak{M}_{\chi,R}^{\text{adj}})$$

is defined by the first-variation identity

$$(5.1) \quad \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} \mathcal{W}_{\chi,R}[V + \varepsilon\varphi, P + \varepsilon\psi, \rho_\varepsilon](s; \sigma) = \int_{B_R} \chi^2 \rho \left(\mathfrak{M}_{\chi,R}^{\text{vel}} \cdot \varphi + \mathfrak{M}_{\chi,R}^{\text{prs}} \psi + \mathfrak{M}_{\chi,R}^{\text{adj}} \eta \right) dy + \mathcal{B}_{\chi,R}[\varphi, \psi, \eta],$$

for every divergence-free $\varphi \in C_c^\infty(B_R; \mathbb{R}^3)$, every mean-free $\psi \in C_c^\infty(B_R)$, and every compactly supported $\eta \in C_c^\infty(B_R)$ satisfying the normalization constraint

$$\int_{B_R} \chi^2 \eta \rho dy = 0,$$

where $\rho_\varepsilon = (4\pi\theta)^{-3/2}e^{-(f+\varepsilon\eta)}$. The term $\mathcal{B}_{\chi,R}$ collects the cutoff, pressure-gauge, and boundary terms created by localization.

Proposition 5.2 (Principal part of the Euler–Lagrange system). *In the notation of Theorem 5.1, the three components of $\mathfrak{G}_{\chi,R}$ have the following principal parts:*

$$(5.2) \quad \mathfrak{M}_{\chi,R}^{\text{vel}} = -2\theta\Delta V + 3\theta|V|V + \theta\nabla\tilde{P} + \text{transport and localization terms},$$

$$(5.3) \quad \mathfrak{M}_{\chi,R}^{\text{prs}} = \frac{3}{2}\theta|\tilde{P}|^{-1/2}\tilde{P} + \text{Calderón–Zygmund pressure-correction terms},$$

and

$$(5.4) \quad \mathfrak{M}_{\chi,R}^{\text{adj}} = -2\theta\Delta f + \theta|\nabla f|^2 + f - 3 + \text{transport and localization terms}.$$

In particular, if the local error term $\mathcal{B}_{\chi,R}$ vanishes in the limit, then the zero-gradient condition

$$\mathfrak{G}_{\chi,R}(V, P, f) = 0$$

is equivalent to a weighted stationary balance among diffusion, nonlinear transport, mean-free pressure, and adjoint concentration.

Proof. Differentiate the terms in (3.5) with respect to the three constrained variables V , P , and f . The variation of the enstrophy term contributes the Laplacian in (5.2); the variation of the local scale-critical mass contributes the cubic term $3\theta|V|V$; the mean-free pressure term contributes (5.3); and the Fisher information together with the linear f -term contributes (5.4). The represented drift $a(s)y + b(s) - V$, the cutoff χ , and the mean-free pressure normalization generate the lower-order transport and localization terms, all of which are grouped into $\mathcal{B}_{\chi,R}$. This yields the stated first-variation structure. $\square$

Definition 5.3 (Zero-gradient branch state). An admissible branch state is called a *zero-gradient branch state* if, for some choice of cutoff χ and normalized adjoint weight ρ , one has

$$\mathfrak{G}_{\chi,R}(V, P, f) = 0$$

and the localization defect satisfies

$$\mathcal{B}_{\chi,R} = 0$$

in the limiting branch topology furnished by the companion compactness theory.

Theorem 5.4 (Recovery of terminal singular geometries from vanishing gradient). *Let $S \in \mathfrak{S}$ be a zero-gradient branch state in the sense of Theorem 5.3. Then, after applying the rerouting theorems already proved in [4, 3, 5], exactly one of the following alternatives holds.*

- (1) **Compact single-core balance.** *The concentration remains localized in a single compact core, the dissipation defect vanishes, and no splitting or tail-loss term survives.*
- (2) **Multiple-core balance.** *The branch decomposes into several nontrivial concentration cores whose local interaction defects vanish after decoupling, corresponding to the multibubble and cascade rows.*
- (3) **Rotational or stationary-hull balance.** *The branch is stationary modulo the rotational or time-translation symmetries isolated in the residual Type I paper.*
- (4) **Scale-cascade balance.** *The vanishing of the entropy gradient is achieved through an exact balance of scale transport, corrected monotonicity, and carrier-routing terms.*
- (5) **Diffuse or tail-defect balance.** *No compact core remains, and the residual balance is carried by diffuse, exterior, or critical-tail concentration.*

Conversely, every terminal model branch isolated in the companion state-space decomposition is represented by a zero-gradient branch state for a suitable choice of localization and adjoint weight.

Proof. The complete state-space decomposition from [4, 3, 5] separates every branch into compact single-core, multiple-core, rotational or stationary-hull, scale-routing, and diffuse or tail-defect regimes, together with intermediate routing rows. The intermediate rows are precisely the rows in which either the localization defect $\mathcal{B}_{\chi,R}$ is nonzero or one of the three Euler–Lagrange residuals in Theorem 5.1 remains active. Since both vanish for a zero-gradient branch state, none of the intermediate routing rows can persist. The only remaining possibilities are the five terminal balance classes listed above.

For the converse statement, fix a terminal model branch obtained from the companion compactness theory. By construction, such a branch is already past all rerouting and defect-producing stages. Choosing χ supported on the relevant compact region and letting ρ solve the normalized adjoint equation on the limiting represented branch gives a localized entropy for which the branch satisfies the stationary balance from Theorem 5.2. Hence it is a zero-gradient branch state. $\square$

Corollary 5.5 (Zero-gradient states and terminal model classes). *The set of zero-gradient branch states coincides, modulo the rerouting theorems of [4, 3, 5], with the union of the five terminal model classes.*

Proof. This is the combination of the forward and converse statements in Theorem 5.4. $\square$

6. ENTROPY-CRITICAL LIMITS AND THE FIVE TERMINAL MODEL CLASSES

Definition 6.1 (Entropy-critical limit). An admissible branch state $S_\infty \in \mathfrak{S}$ is called *entropy-critical* if there exist an admissible singular branch $S_n \in \mathfrak{S}$, times $s_n \uparrow \sigma$, and cutoffs χ such that:

- (i) $S_n \rightarrow S_\infty$ in the companion compactness topology;
- (ii) $\mathcal{W}_{\chi,R}[S_n](s_n; \sigma)$ converges;
- (iii) the dissipation and error terms in (4.1) satisfy

$$\mathcal{D}_{\chi,R}(s_n; \sigma) \rightarrow 0, \quad \mathcal{E}_{\chi,R}(s_n - \delta, s_n; \sigma) \rightarrow 0$$

for every fixed $\delta > 0$ with $s_n - \delta > -R^2$.

Proposition 6.2 (Entropy compactification). *Every hypothetical singular branch in the companion state space has an entropy-critical subsequential limit.*

Proof. By Theorem 4.2, the localized entropy has a finite limiting value along terminal sequences. The compactness theorems in [4, 3, 5] provide subsequential limits in the corresponding state-space topologies: bounded centered ancient profiles on the Type I side, and represented local

limits on the Type II side. Since the integrated dissipation in (4.1) is nonnegative and the entropy converges, a standard telescoping argument yields a sequence along which the dissipation tends to zero. The routing defects are controlled by the companion closure theorems, so one may pass to a further subsequence for which the error term also vanishes. The resulting limit is entropy-critical. $\square$

The point of the entropy formalism is that not every row in the companion decision tree can be entropy-critical. Many rows are routing mechanisms rather than terminal geometries. Once the error term vanishes, only the terminal classes remain, and by Theorem 5.5 these are exactly the zero-gradient branch states.

Theorem 6.3 (Classification of entropy-critical limits). *Every entropy-critical limit belongs, after the reroutings already proved in [4, 3, 5], to one of the following five terminal model classes:*

- (1) **Compact single-core class.** *This includes the direct Type I bounded ancient branches and the Type II compact single-core branches with retained concentration.*
- (2) **Multiple-core class.** *This includes multibubble, separated-profile, and same-point cascade configurations.*
- (3) **Rotational or stationary-hull class.** *This is the residual Type I branch governed by rotational relative equilibria and stationary-hull compactifications.*
- (4) **Scale-cascade class.** *This includes scale-collapse, scale-rigid, cost-divergence, and canonical carrier-routing branches.*
- (5) **Diffuse or tail-defect class.** *This includes critical-tail defects, diffuse terminal defects, terminal nonactive branches, and exterior-loss mechanisms.*

Proof. Start from the complete state-space decompositions proved in the companion papers. The direct Type I classes and the compact single-core Type II class are already terminal compactness mechanisms, so they form the compact single-core class. The multibubble, separated-profile, and same-point cascade rows are all instances of multiple localized concentration and therefore form the multiple-core class. The rotational and stationary-hull rows are terminal residual compactifications in [3], so they form the third class. The scale-collapse, scale-rigid, cost-divergence, and carrier-routing rows are all driven by noncompact behavior in scale and therefore form the scale-cascade class. Finally, the critical-tail, diffuse terminal, nonactive, and exterior-loss rows are the remaining classes in which concentration fails to remain in a compact core, so they form the diffuse or tail-defect class.

What remains to check is that no purely routing row can be entropy-critical. But by construction the routing rows are precisely the rows that contribute to the error term $\mathcal{E}_{\chi,R}$ in Theorem 4.1. If such a row remained active in the limit, then $\mathcal{E}_{\chi,R}$ would not vanish in the sense of Theorem 6.1. Therefore an entropy-critical limit must lie in one of the five terminal model classes above. $\square$

The classification can be summarized as follows.

Terminal class	Manifestations in the companion state space	Closed in
Compact single-core	Direct Type I bounded ancient branches; Type II compact single-core and retained compact-test branches	[4, 5]
Multiple-core	Multibubble, separated-profile, and same-point cascade branches	[5]
Rotational/stationary- hull	Rotational relative equilibria and stationary-hull residual branches	[3]
Scale-cascade	Scale-collapse, scale-rigid, cost-divergence, and carrier-routing branches	[5]
Diffuse/tail-defect	Critical-tail defects, diffuse terminal defects, terminal nonactive branches, and exterior concentration	[3, 5]

7. REDUCTION TO THE COMPANION EXCLUSION THEOREMS

The entropy formulation becomes effective only after it is paired with the closure results already proved in the companion papers.

Theorem 7.1 (Elimination of the terminal classes). *Each of the five terminal model classes in Theorem 6.3 is empty. More precisely:*

- (i) *the compact single-core class is excluded by the direct Type I Liouville closures of [4] and the compact single-core and retained compact-test closures of [5];*
- (ii) *the multiple-core class is excluded by the multibubble, cascade, separated-profile, and terminal profile reductions of [5];*
- (iii) *the rotational or stationary-hull class is excluded by the residual closures of [3];*
- (iv) *the scale-cascade class is excluded by the scale-collapse, scale-rigid, corrected-monotonicity, and canonical carrier-routing theorems of [5];*
- (v) *the diffuse or tail-defect class is excluded by the critical-tail and diffuse-defect analysis of [3] together with the exterior and terminal-routing closures of [5].*

Proof. This is a direct restatement of the closure results recorded in Theorem 2.1. The point is not that the entropy formulation adds new exclusion lemmas; rather, it shows that the companion exclusion lemmas are exactly the local mechanisms needed to eliminate all entropy-critical limits. $\square$

Combining Theorems 6.2, 6.3 and 7.1 with the companion state-space reduction yields the entropy version of the three-paper proof strategy.

Corollary 7.2 (Entropy reformulation of the local exclusion program). *Within the framework of [4, 3, 5], every hypothetical singular branch loses entropy until it either enters a rerouting row already controlled by local estimates or converges to a terminal model class. Since every terminal model class is empty, no singular branch remains.*

Remark 7.3. The corollary should be read as an entropy-level summary of the companion papers, not as a replacement for them. The local pressure decomposition, carrier routing, terminal profile analysis, and residual compactification are the substantive PDE content. The entropy formalism shows that these arguments fit together as a finite compactification mechanism.

8. COMPARISON WITH HAMILTON–PERELMAN MONOTONICITY

The analogy with Hamilton and Perelman is useful but should not be overstated. In Ricci flow, the $\mathcal{W}$ -functional is global and geometric, with the scalar curvature appearing explicitly [12, 20].

In the present Navier–Stokes setting, there is no corresponding global geometric functional available at the level of the companion proofs. The pressure is nonlocal, and the branch changes are genuinely local. For that reason, the functional $\mathcal{W}$ is localized from the outset.

Nevertheless, the two theories share a common pattern.

- (1) Both use a backward adjoint weight to measure concentration near a candidate terminal time.
- (2) Both produce a monotone quantity whose defect is a sum of nonnegative dissipation and explicit local error terms.
- (3) In both settings, a critical point of the monotone quantity represents a terminal model geometry.

The decisive difference is that, in the present paper, the critical classes are not derived from an abstract classification theorem. They are read off from the local state-space decomposition already established in the companion manuscripts.

9. CONCLUDING REMARKS

The companion papers already provide a complete local routing architecture. The point of the present manuscript is to show that this architecture admits a coherent entropy interpretation in standard PDE language. Once the local enstrophy, mean-free pressure, scale-critical mass, and backward adjoint weight are assembled into $\mathcal{W}$, the decision tree may be read as a compactness statement for the entropy flow: every singular branch either dissipates into a previously closed local row or converges to one of five terminal model classes.

This viewpoint suggests two further directions. First, it separates the purely local analytic input from the compactification mechanism, which may be useful in formalization or in alternative singularity analyses. Second, it clarifies the role of the local pressure decomposition: the pressure does not obstruct a monotone quantity once it is measured in the local mean-free norms naturally produced by the companion proofs.

REFERENCES

- [1] D. Albritton and T. Barker, *On local Type I singularities of the Navier–Stokes equations and Liouville theorems*, J. Math. Fluid Mech. **21** (2019), Article 43.
- [2] V. I. Arnold, *Sur la géométrie différentielle des groupes de Lie de dimension infinie et ses applications à l’hydrodynamique des fluides parfaits*, Ann. Inst. Fourier (Grenoble) **16** (1966), no. 1, 319–361.
- [3] G. Duran Ballester, *Local Exclusion of Residual Type I Navier–Stokes Seregin Limits*, companion manuscript, 2026.
- [4] G. Duran Ballester, *Conditional Local Type I Navier–Stokes Reduction: Concentration, Seregin Limits, and Liouville-Class Closures*, companion manuscript, 2026.
- [5] G. Duran Ballester, *Type II Regularity for Navier–Stokes*, companion manuscript, 2026.
- [6] A. P. Calderón and A. Zygmund, *On the existence of certain singular integrals*, Acta Math. **88** (1952), 85–139.
- [7] L. Caffarelli, R. Kohn, and L. Nirenberg, *Partial regularity of suitable weak solutions of the Navier–Stokes equations*, Comm. Pure Appl. Math. **35** (1982), no. 6, 771–831.
- [8] P. Constantin and C. Foias, *Navier–Stokes Equations*, University of Chicago Press, Chicago, 1988.
- [9] D. G. Ebin and J. Marsden, *Groups of diffeomorphisms and the motion of an incompressible fluid*, Ann. of Math. (2) **92** (1970), no. 1, 102–163.
- [10] L. Escauriaza, G. A. Seregin, and V. Šverák, *$L_{3,\infty}$ -solutions of the Navier–Stokes equations and backward uniqueness*, Russian Math. Surveys **58** (2003), no. 2, 211–250.
- [11] S. Gustafson, K. Kang, and T.-P. Tsai, *Interior regularity criteria for suitable weak solutions of the Navier–Stokes equations*, Comm. Math. Phys. **273** (2007), no. 1, 161–176.
- [12] R. S. Hamilton, *Three-manifolds with positive Ricci curvature*, J. Differential Geom. **17** (1982), no. 2, 255–306.
- [13] T. Kato, *Strong L^p -solutions of the Navier–Stokes equation in $\mathbb{R}^m$, with applications to weak solutions*, Math. Z. **187** (1984), no. 4, 471–480.
- [14] G. Koch, N. Nadirashvili, G. Seregin, and V. Šverák, *Liouville theorems for the Navier–Stokes equations and applications*, Acta Math. **203** (2009), no. 1, 83–105.

- [15] O. A. Ladyzhenskaya, *The Mathematical Theory of Viscous Incompressible Flow*, Gordon and Breach, New York, 1969.
 - [16] P. G. Lemarié-Rieusset, *Recent Developments in the Navier–Stokes Problem*, Chapman & Hall/CRC, Boca Raton, FL, 2002.
 - [17] J. Leray, *Sur le mouvement d'un liquide visqueux emplissant l'espace*, Acta Math. **63** (1934), no. 1, 193–248.
 - [18] F.-H. Lin, *A new proof of the Caffarelli–Kohn–Nirenberg theorem*, Comm. Pure Appl. Math. **51** (1998), no. 3, 241–257.
 - [19] J. Nečas, M. Růžička, and V. Šverák, *On Leray's self-similar solutions of the Navier–Stokes equations*, Acta Math. **176** (1996), no. 2, 283–294.
 - [20] G. Perelman, *The entropy formula for the Ricci flow and its geometric applications*, arXiv:math/0211159.
 - [21] V. Scheffer, *Partial regularity of solutions to the Navier–Stokes equations*, Pacific J. Math. **66** (1976), no. 2, 535–552.
 - [22] G. Seregin, *A certain necessary condition of potential blow up for Navier–Stokes equations*, Comm. Math. Phys. **312** (2012), no. 3, 833–845.
 - [23] E. M. Stein, *Singular Integrals and Differentiability Properties of Functions*, Princeton University Press, Princeton, NJ, 1970.
 - [24] R. Temam, *Navier–Stokes Equations: Theory and Numerical Analysis*, North-Holland, Amsterdam, 1977.
- Email address: guillem@fragile.tech